EPL Player Segmentation Analysis

Unsupervised Clustering & Dimensionality Reduction on 2020–21 English Premier League Player Data

Project Domain:	Sports Analytics & Unsupervised Machine Learning
Dataset:	Epl_data.csv — 476 English Premier League players, 10 numeric performance features
Core Methods:	StandardScaler, PCA, K-Means Clustering
Status:	Segmentation Validated — 3 Clusters Identified

EXECUTIVE SUMMARY

This report documents an unsupervised segmentation analysis of 476 English Premier League (EPL) players from the 2020–21 season. Ten numeric performance features were standardized, reduced via Principal Component Analysis (PCA), and clustered using K-Means to identify natural player archetypes. Three statistically distinct player segments were recovered, differentiated primarily by attacking output and playing time rather than by nominal squad position.

1. DATA & PREPROCESSING PIPELINE

The source dataset contains 476 players across 13 columns: identifying fields (Player_Name, Club, Position) and 10 numeric features (Goals_Scored, Assists, Total_Points, Minutes, Goals_Conceded, Creativity, Influence, Threat, Bonus, Clean_Sheets). No duplicate records were found. Because these features span very different raw scales (Minutes in the thousands vs. Goals_Scored in single digits), all 10 numeric columns were standardized to zero mean and unit variance using **StandardScaler** prior to any downstream modeling.

2. DIMENSIONALITY REDUCTION (PCA)

PCA was fit across all 10 standardized features to evaluate the underlying variance structure of the data.

Components Retained	Cumulative Variance Explained
1	72.2%
2	85.8%
3	91.7%
4	94.8%
5	97.1%

The first three principal components jointly explain **91.7%** of total variance — comfortably above the conventional 90% retention threshold. Three components were therefore retained as the basis for clustering, balancing dimensionality reduction against information loss.

3. DETERMINING THE OPTIMAL NUMBER OF CLUSTERS

K-Means was evaluated for k = 2 through k = 10 on the 3-component PCA-transformed data, using both inertia (elbow method) and silhouette score as selection criteria.

k	Inertia	Silhouette Score
2	1946.16	0.523
3	1193.19	0.514
4	928.82	0.468
5	736.52	0.478
6	603.98	0.487

Silhouette score peaks at k = 2 (0.523), with k = 3 close behind (0.514) before dropping off more steeply from k = 4 onward. **k = 3** was selected: it retains a comparably strong silhouette score while producing more actionable, distinct player segments than a 2-cluster split would allow.

4. CLUSTERING RESULTS

Cluster	Players	Share of Dataset
0	63	13.2%
1	248	52.1%
2	165	34.7%

Cluster 0 — "Attacking Threats" (63 players, 13.2%)

Highest standardized (z-score) values on Goals_Scored (+1.98), Threat (+1.97), Assists (+1.85), Bonus (+1.85) and Creativity (+1.71). Composition: 54% Midfielders, 32% Forwards, 14% Defenders, 0% Goalkeepers. Raw averages: 8.7 goals, 6.8 assists, 142.1 total points per player — the standout attacking and point-scoring group.

Cluster 1 — "Fringe / Low-Minute Players" (248 players, 52.1%)

Lowest standardized values across nearly every feature, most notably Minutes (-0.82), Influence (-0.80), and Total_Points (-0.79). A mixed-position group (38% Defenders, 37% Midfielders, 15% Forwards, 11% Goalkeepers) unified by limited playing time rather than position — largely bench, rotation, or squad-depth players. Raw average: 18.0 total points per player.

Cluster 2 — "Defensive Regulars" (165 players, 34.7%)

Highest standardized values on Minutes (+0.83), Goals_Conceded (+0.80), Clean_Sheets (+0.76), and Influence (+0.68), with attacking metrics near the population average. Composition: 42% Defenders, 42% Midfielders, 10% Goalkeepers, 5% Forwards. This group plays consistent minutes and accumulates defensive counting statistics — regular starters, concentrated at the back but not exclusively defenders.

Key Technical Insights

• **Segmentation is minutes-driven, not position-driven:** the dominant separator between clusters is playing time and attacking output, not the nominal Position field — Clusters 1 and 2 each mix Defenders and Midfielders roughly evenly.

• **Profiling methodology correction:** an earlier profiling pass compared raw feature means and inadvertently included the engineered Cluster label itself as an input column, causing it to appear as a top "differentiating feature." This was corrected by excluding the Cluster column and using standardized z-score means, which control for the very different raw scales across features.

- **k = 2 vs. k = 3 trade-off:** silhouette score alone marginally favors k = 2 (0.523 vs. 0.514), so the choice of k = 3 should be explicitly justified in terms of interpretability, not just the silhouette metric.

5. CONCLUSIONS & RECOMMENDATIONS

PCA followed by K-Means (k = 3) on standardized EPL player features produces three interpretable, performance-based player segments that cut across the nominal Position field: a small elite attacking group, a large low-minutes fringe group, and a mid-sized defensively-oriented regular-starter group.

Recommendations:

- Explicitly justify k = 3 over k = 2 using both the elbow plot and silhouette score, since k = 2 technically edges out k = 3 on silhouette score alone.
- Keep the corrected profiling approach (exclude the Cluster column; use standardized z-score means) when reporting differentiating features.
- Consider per-90-minutes normalized features (e.g., goals per 90) to separate playing quality from playing time, since Cluster 1 is currently defined largely by low minutes rather than low underlying quality.
- Cross-tabulate clusters against Position and Club to confirm whether segments reflect tactical roles or simply playing-time tiers.

Final Project Conclusion

The clustering pipeline (StandardScaler → PCA (3 components, 91.7% variance) → K-Means, k = 3) is methodologically sound and produces a stable, interpretable three-segment player typology. The principal refinement needed before final submission is correcting the cluster-profiling step to exclude the engineered Cluster column, as documented above.